



Koneru Lakshmaiah Education Foundation

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Course Name: MATHEMATICS FOR DATA SCIENCE & ANALYTICS (25MT1306E)

CO-3: Session-18: Normal Distributions with related properties

Case Study 1: Industrial Production

In a manufacturing unit, the diameter of steel bolts produced is normally distributed with a mean of 20 mm and a standard deviation of 1.5 mm. Due to strict quality requirements, bolts must fall within a specific tolerance range to be accepted. Any bolt outside this range is rejected and leads to financial loss. The company produces thousands of bolts daily and continuously monitors quality performance.

Tasks:

- What proportion of bolts have diameters between 18.5 mm and 21.5 mm?
- If bolts with diameters less than 17 mm or greater than 23 mm are rejected, what is the rejection rate?
- If 5,000 bolts are produced daily, estimate how many are acceptable.
- Suggest how reducing variability will impact rejection rate and explain mathematically.

Case Study 2: Academic Performance

In a competitive examination, student scores are normally distributed with a mean of 65 marks and a standard deviation of 8 marks. The examination board uses statistical methods to determine performance categories such as average, above average, and exceptional. Scholarships are awarded to top-performing students based on percentile ranking.

Tasks

- What percentage of students score above 75 marks?
- Determine the cutoff score for the top 5% of students.
- If 10,000 students appeared, estimate how many scored between 60 and 70 marks.
- Explain how increasing the mean score would affect percentile rankings without changing standard deviation.

Case Study 3: Health Monitoring

A health organization studies the cholesterol levels of adults in a city, which are normally distributed with a mean of 200 mg/dL and a standard deviation of 25 mg/dL. Individuals with cholesterol levels above a certain threshold are considered at risk and are advised to have medical intervention. Public health policies are designed based on such statistical distributions.

Tasks

- What proportion of individuals have cholesterol levels between 180 and 220 mg/dL?
- Determine the cholesterol level above which 10% of individuals fall.
- If 3,000 people are tested, estimate how many are in the high-risk category (top 10%).
- Discuss how a shift in the mean cholesterol level impacts the overall health risk in the population.

Concept of the Normal Distribution

The normal distribution (or normal random variable) is one of the most important continuous probability distributions in statistics. It serves as the foundation for understanding the standard normal distribution. Its characteristic bell-shaped curve effectively represents many phenomena observed in science, business, and the natural world.

Another related concept is the normal random variable, which is a continuous random variable X that follows a bell-shaped distribution. Its probability distribution is completely determined by two parameters: μ (the population mean) and σ (the population standard deviation). Accordingly, the density of X is denoted by $n(x; \mu, \sigma)$.

The probability density function of a normal random variable is given by:

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

This expression defines the bell-shaped curve for a normal distribution with mean μ and standard deviation σ .

The **standard normal distribution**, also known as the **Z-distribution**, is a special case of the normal distribution in which the mean (μ) is 0 and the standard deviation (σ) is 1.

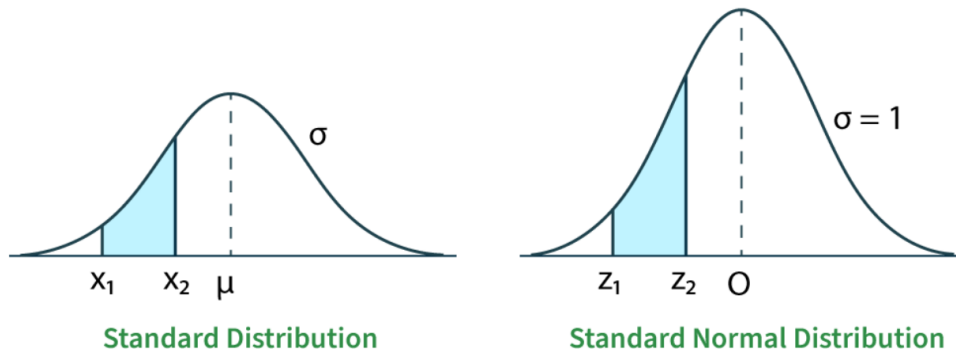
If we define the standardized variable as

$$Z = \frac{x - \mu}{\sigma},$$

then for $\mu = 0$ and $\sigma = 1$, the probability density function becomes:

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, -\infty < x < \infty$$

Thus, a normal random variable with mean 0 and variance 1 ($\sigma^2 = 1$) is called a standard normal distribution. This distinguishes it from the general normal distribution, which can have any mean and standard deviation.



Application of Standard Normal Distribution

Standard Normal Distribution has a wide range of applications and usage in several fields. Here are some important applications:

Hypothesis Testing: Performing Z-tests and constructing confidence intervals.

Probability Calculations: Determining standard normal distribution density probabilities and areas under the curve.

Data Analysis: Standardizing scores (z-scores) and making suitable analysis.

Quality Control: Monitoring processes using control charts.

Risk Management: Calculating financial risks like Value at Risk (VaR).

Machine Learning: Normalizing features to improve algorithm performance.

Modeling Phenomena: Analyzing real-world data in natural and social sciences.

Survey Analysis: Estimating and comparing population proportions.

Psychometrics: Using standard scores in psychological assessments.

Operations Research: Optimizing inventory management based on demand modeling.

Characteristics of Standard Normal Distribution

Mean: The mean (average) is 0 that is symbolically represented as $\mu = 0$.

Standard Deviation: The standard deviation is 1 that is symbolically represented as $\sigma = 1$.

Symmetry: It is symmetric around the mean ($\mu = 0$).

Bell-Shaped Curve: The graph is bell-shaped, that means most values cluster around the mean ($\mu = 0$).

Total Area Under Curve: The total area under the curve is 1, representing the total probability.

68-95-99.7 Rule: Approximately 68% of data falls within 1 standard deviation of the mean, 95% within 2 standard deviations, and 99.7% within 3 standard deviations.

Asymptotic: The tails of the distribution approach, but never touch, the horizontal axis.

Unimodal: It has a single peak at the mean ($\mu = 0$).

Standard Scores (Z-Scores): Any normal distribution can be transformed into the standard normal distribution using z-scores where $z = (x - \mu)/\sigma$.

Case Study 1: Industrial Production

In a manufacturing unit, the diameter of steel bolts produced is normally distributed with a mean of 20 mm and a standard deviation of 1.5 mm. Due to strict quality requirements, bolts must fall

within a specific tolerance range to be accepted. Any bolt outside this range is rejected and leads to financial loss. The company produces thousands of bolts daily and continuously monitors quality performance.

Tasks:

- What proportion of bolts have diameters between 18.5 mm and 21.5 mm?
- If bolts with diameters less than 17 mm or greater than 23 mm are rejected, what is the rejection rate?
- If 5,000 bolts are produced daily, estimate how many are acceptable.
- Suggest how reducing variability will impact rejection rate and explain mathematically.

Solution:

Mean $\mu = 20\text{mm}$ Standard deviation $\sigma = 1.5\text{mm}$ We use the standard normal transformation: $Z = \frac{X-\mu}{\sigma}$	Acceptable bolts out of 5,000 Acceptable proportion: $1 - 0.0455 = 0.9545$ Number of acceptable bolts: $5000 \times 0.9545 = 4772.5 \approx 4773$
Given that the proportion of bolts between 18.5 mm and 21.5 mm. Convert to Z-scores: $Z_1 = \frac{18.5 - 20}{1.5} = -1, Z_2 = \frac{21.5 - 20}{1.5} = 1$ So, $P(18.5 < X < 21.5) = P(-1 < Z < 1)$ From standard normal tables: $P(-1 < Z < 1) = 0.6826$	Effect of reducing variability (standard deviation). To reduce rejection rate by below 1%: We need: $P(Z > k) < 0.01 \Rightarrow k \approx 2.575$. Tolerance limits are fixed: 17 and 23 So, $k = \frac{3}{\sigma}$ and Set: $\frac{3}{\sigma} = 2.575$ $\sigma = \frac{3}{2.575} \approx 1.165$
Rejection rate ($X < 17$ or $X > 23$) $Z = \frac{17 - 20}{1.5} = -2, Z = \frac{23 - 20}{1.5} = 2$ $P(X < 17 \text{ or } X > 23) = P(Z > 2)$ From tables: $P(Z > 2) = 0.0455$	Conclusion: Current $\sigma = 1.5$ and required $\sigma \approx 1.165$ So, standard deviation must decrease to reduce defects. Interpretation: Lower variability $\rightarrow$ data more concentrated around mean Fewer extreme values $\rightarrow$ fewer rejected bolts
Summary: 68.26% within tolerance 4.55% rejected ~4,773 acceptable bolts - Reduce σ from 1.5 to about 1.165 to achieve <1% rejection	

Standard Normal Distribution Table

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Case Study 4: Customer Service Efficiency

A telecom company records the time taken to resolve customer complaints, which follows a normal distribution with a mean of 15 minutes and a standard deviation of 3 minutes. Customer satisfaction is closely linked to resolution time, and delays beyond a certain threshold are considered poor service.

Tasks

- What is the probability that a complaint is resolved in less than 12 minutes?
- Find the probability that resolution time exceeds 20 minutes.
- If the company handles 2,000 complaints daily, estimate how many take more than 18 minutes.
- Explain how reducing the mean resolution time affects service efficiency and customer satisfaction.

Case Study 5: Product Reliability

The lifetime of LED bulbs manufactured by a company follows a normal distribution with a mean of 1,200 hours and a standard deviation of 150 hours. Customers expect consistent performance, and the company uses statistical analysis to estimate warranty claims and product reliability.

Tasks

- What proportion of bulbs last more than 1,350 hours?
- Determine the lifetime below which 20% of bulbs fail.
- If 6,000 bulbs are sold, estimate how many last between 1,000 and 1,400 hours.
- Analyze how reducing standard deviation improves product reliability and reduces warranty costs.

Case Study 6: Traffic Flow Analysis

Traffic speeds on a major highway are normally distributed with a mean of 70 km/h and a standard deviation of 12 km/h. Authorities monitor speed patterns to enforce safety regulations and reduce accidents. Speed limits are imposed based on statistical analysis of vehicle speeds.

Tasks

- What proportion of vehicles travel below 60 km/h?
- Find the speed exceeded by only 15% of vehicles.
- If 4,000 vehicles pass a checkpoint daily, estimate how many exceed 85 km/h.
- Discuss how reducing the average speed affects accident probability and traffic safety.

MCQs on Normal Distribution

- The mean and standard deviation of a standard normal distribution are:
A) $\mu = 1, \sigma = 0$ B) $\mu = 0, \sigma = 1$ C) $\mu = 1, \sigma = 1$ D) $\mu = 0, \sigma = 0$
- The total area under the normal distribution curve is:
A) 0 B) 0.5 C) 1 D) ∞
- The normal distribution curve is:
A) Asymmetrical B) Skewed left C) Symmetrical D) Discrete
- In a normal distribution, approximately 68% of data lies within:
A) $\mu \pm 1\sigma$ B) $\mu \pm 2\sigma$ C) $\mu \pm 3\sigma$ D) $\mu \pm 4\sigma$
- If $Z = 1.0$, then $P(Z \leq 1.0) \approx$:
A) 0.1587 B) 0.5000 C) 0.8413 D) 0.9772
- If $Z = -1.0$, then $P(Z \leq -1.0) \approx$:

- A) 0.8413 B) 0.1587 C) 0.9772 D) 0.5000
7. The Z-score formula is:
 A) $Z = \frac{X+\mu}{\sigma}$ B) $Z = \frac{X-\mu}{\sigma}$ C) $Z = \frac{\mu-X}{\sigma}$ D) $Z = X\mu\sigma$
8. If the standard deviation increases, the normal curve becomes:
 A) Narrower B) Flatter and wider C) Steeper D) Unchanged
9. The probability that a standard normal variable takes a value exactly equal to a point is:
 A) 1 B) 0.5 C) 0 D) ∞
10. For a normal distribution, mean, median, and mode are:
 A) Different B) Equal C) Random D) Undefined

Answer Key:

1-B 2-C 3-C 4-A 5-C 6-B 7-B 8-B 9-C 10-B

Key Insights:

1. Bell-Shaped and Symmetrical
 - The normal distribution curve is bell-shaped and perfectly symmetrical about the mean.
 - Left side = right side (mirror image).
2. Mean = Median = Mode
 - All three measures of central tendency are equal:

$$\mu = \text{Median} = \text{Mode}$$
3. Total Area = 1
 - The total area under the curve is 1, representing total probability.
 - 50% of data lies on each side of the mean.
4. Empirical Rule (68–95–99.7 Rule)
 - About 68% of data lies within $\mu \pm 1\sigma$
 - About 95% lies within $\mu \pm 2\sigma$
 - About 99.7% lies within $\mu \pm 3\sigma$
5. Standard Normal Distribution (Z-Distribution)
 - Special case where:

$$\mu = 0, \sigma = 1$$

Any normal variable can be converted using:

$$Z = \frac{X - \mu}{\sigma}$$
6. Role of Mean and Standard Deviation
 - Mean (μ): determines the center of the curve
 - Standard deviation (σ): determines spread
 - Larger $\sigma \rightarrow$ wider, flatter curve
 - Smaller $\sigma \rightarrow$ narrower, taller curve
7. Asymptotic Nature
 - The curve never touches the horizontal axis, only approaches it.
8. Continuous Distribution
 - It is a continuous probability distribution
 - Probability at a single point is zero, only intervals matter.
9. Standardization Simplifies Problems
 - Converting to Z-scores allows use of Z-tables for probability calculation.
10. Wide Real-Life Applications

Used in:

- Quality control (manufacturing)
- Exam score analysis
- Medical measurements
- Financial modeling
- Natural phenomena